

CO4 - AT2 - Coding Golf Assessment

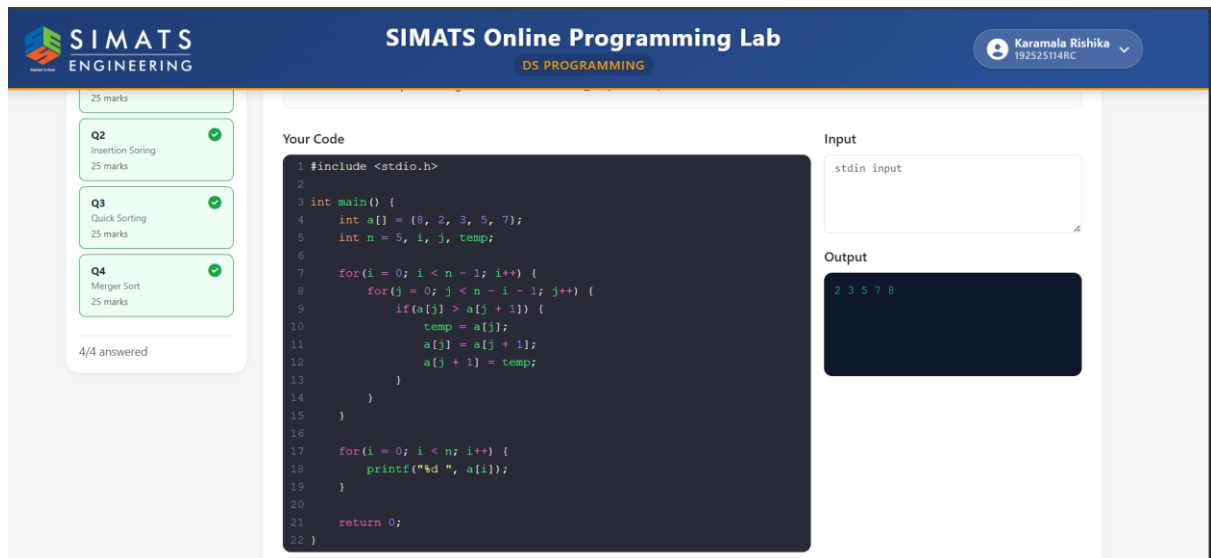
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Course Code:CSA0303

Course Name: Data Structures

1.

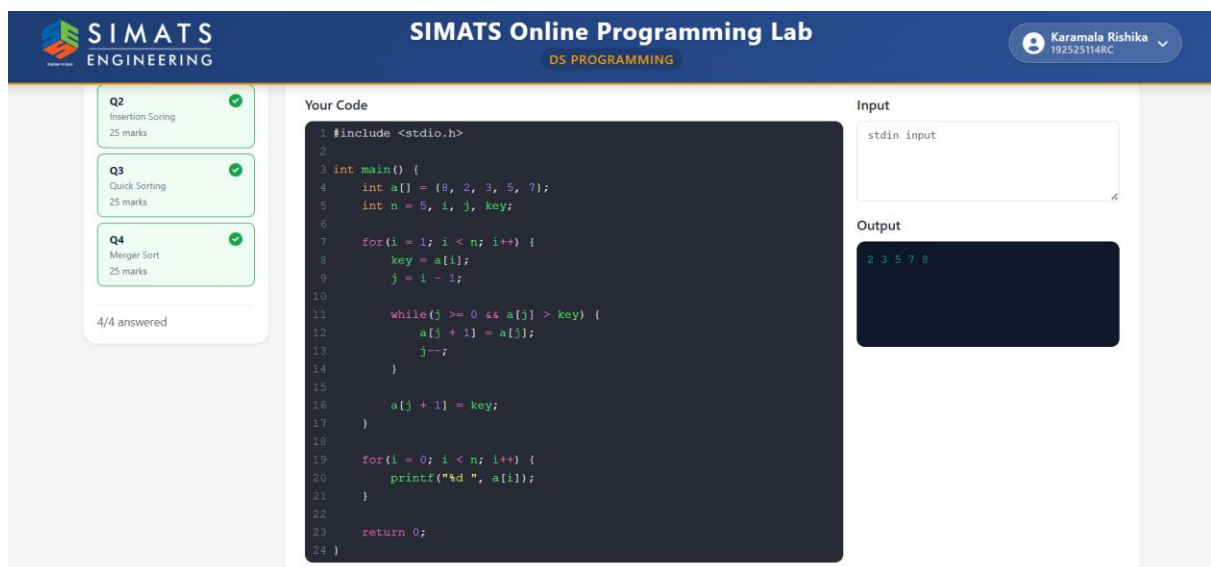


The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS ENGINEERING logo, the lab name "SIMATS Online Programming Lab", the course "DS PROGRAMMING", and the user "Karamala Rishika" with ID "192525114RC". On the left, a sidebar lists three questions: Q2 (Insertion Sorting, 25 marks), Q3 (Quick Sorting, 25 marks), and Q4 (Merger Sort, 25 marks). Q2 is selected and marked as answered. The main area displays the code for Insertion Sorting. The code is as follows:

```
1 #include <stdio.h>
2
3 int main() {
4     int a[] = {8, 2, 3, 5, 7};
5     int n = 5, i, j, temp;
6
7     for(i = 0; i < n - 1; i++) {
8         for(j = 0; j < n - i - 1; j++) {
9             if(a[j] > a[j + 1]) {
10                 temp = a[j];
11                 a[j] = a[j + 1];
12                 a[j + 1] = temp;
13             }
14         }
15     }
16
17     for(i = 0; i < n; i++) {
18         printf("%d ", a[i]);
19     }
20
21     return 0;
22 }
```

The input field is labeled "stdin input" and is empty. The output field shows the result: "2 3 5 7 8".

2.



The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS ENGINEERING logo, the lab name "SIMATS Online Programming Lab", the course "DS PROGRAMMING", and the user "Karamala Rishika" with ID "192525114RC". On the left, a sidebar lists three questions: Q2 (Insertion Sorting, 25 marks), Q3 (Quick Sorting, 25 marks), and Q4 (Merger Sort, 25 marks). Q3 is selected and marked as answered. The main area displays the code for Quick Sorting. The code is as follows:

```
1 #include <stdio.h>
2
3 int main() {
4     int a[] = {8, 2, 3, 5, 7};
5     int n = 5, i, j, key;
6
7     for(i = 1; i < n; i++) {
8         key = a[i];
9         j = i - 1;
10
11         while(j >= 0 && a[j] > key) {
12             a[j + 1] = a[j];
13             j--;
14         }
15
16         a[j + 1] = key;
17     }
18
19     for(i = 0; i < n; i++) {
20         printf("%d ", a[i]);
21     }
22
23     return 0;
24 }
```

The input field is labeled "stdin input" and is empty. The output field shows the result: "2 3 5 7 8".

3.

The screenshot displays the SIMATS Online Programming Lab interface. The header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", the subject "DS PROGRAMMING", and a user profile for "Karamala Rishika" with ID "192525114RC". On the left sidebar, two questions are listed: "Q3 Quick Sorting 25 marks" and "Q4 Merger Sort 25 marks", both marked as answered. The main area shows a C++ code editor with a Quick Sort implementation. The code defines a recursive function `quickSort` and a `main` function. The output window on the right shows the sorted array: "1 2 3 5 6 7 8".

```
3 void quickSort(int a[], int low, int high) {
4     int i = low, j = high, pivot = a[(low + high) / 2], temp;
5
6     while (i <= j) {
7         while (a[i] < pivot)
8             i++;
9
10        while (a[j] > pivot)
11            j--;
12
13        if (i <= j) {
14            temp = a[i];
15            a[i] = a[j];
16            a[j] = temp;
17            i++;
18            j--;
19        }
20    }
21
22    if (low < j)
23        quickSort(a, low, j);
24
25    if (i < high)
26        quickSort(a, i, high);
27 }
28
29 int main() {
```

Output

1 2 3 5 6 7 8

4.

The screenshot displays the SIMATS Online Programming Lab interface. The header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", the subject "DS PROGRAMMING", and a user profile for "Karamala Rishika" with ID "192525114RC". On the left sidebar, two questions are listed: "Q3 Quick Sorting 25 marks" and "Q4 Merger Sort 25 marks", both marked as answered. The main area shows a C++ code editor with a Merge Sort implementation. The code defines recursive functions `merge` and `mergeSort`, and a `main` function. The output window on the right shows the sorted array: "1 2 3 4 5 6 7 8".

```
3 void merge(int a[], int l, int m, int r) {
4     int i = l, j = m + 1, k = 0;
5     int temp[20];
6
7     while (i <= m && j <= r) {
8         if (a[i] < a[j])
9             temp[k++] = a[i++];
10        else
11            temp[k++] = a[j++];
12    }
13
14    while (i <= m)
15        temp[k++] = a[i++];
16
17    while (j <= r)
18        temp[k++] = a[j++];
19
20    for (i = l, k = 0; i <= r; i++, k++)
21        a[i] = temp[k];
22 }
23
24 void mergeSort(int a[], int l, int r) {
25     if (l < r) {
26         int m = (l + r) / 2;
27
28         mergeSort(a, l, m);
29         mergeSort(a, m + 1, r);
```

Output

1 2 3 4 5 6 7 8